Housing Allocation, Fertility, and Population

Proposed theoretical amendments

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This note develops the agreed allocation and fertility arguments and records the remaining transition work. The first welfare comparison fixes fertility: its gain comes from reallocating housing services across existing households. Fertility responses are an additional, conditional implication. The financing permission is accepted; the compensation scheme and real reallocation cost remain provisional choices for the author's review. Nothing in this note amends the author-controlled manuscript.

1 Environment

Households and preferences. Each cohort lives for a young and an old period. A young household chooses completed fertility n, nondurable expenditure c, and housing h. Each child requires $\chi > 0$ goods and $\kappa > 0$ housing units, leaving adult goods $x = c - \chi n$ and adult space $s = h - \kappa n$. Lifetime utility is:

$$\log x + \alpha \log s + \vartheta \log n + \beta \left(\log c_2 + \gamma \log h_2 + \omega_B \log e \right). \tag{1}$$

All arguments and preference weights are positive. The estate e is a warm-glow expenditure; it does not determine the next cohort's liquid wealth.

Markets and finance. The fixed housing stock trades at P_t per unit. External lenders supply a one-period bond at price $q \in (0, 1)$, with gross return $1/q$; only housing clears domestically. Rent r_t and property tax $\tau^p P_t$ are payable in next-date goods. Competitive intermediation equates the current value of one period's housing services to the purchase price plus property tax less resale:

$$u_t = q r_t = P_t + q \tau^p P_t - q P_{t+1} > 0. \tag{2}$$

Capital-gains taxation is absent from this core argument and remains a deferred extension. Property taxation is retained.

Access and ownership. A buyer with liquid wealth b finances share $\phi \in (0, 1)$ at purchase and must post $(1 - \phi) P_t h \leq b$ before income and rebates arrive. Separate rental and owner size limits remain as in the underlying model. An old owner can sell housing or retain it for personal use; retained housing enters the estate at death. The planner may finance an additional purchase beyond the private down-payment limit. This permission does not remove real costs or allow the planner to default on lenders.

Population. There are Y_t young and O_t old household decision units. One model child produces $\nu > 0$ entering household units next period. Their laws of motion are:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \qquad O_{t+1} = Y_t. \tag{3}$$

The parameter ν combines the choice of child units, survival, and household formation. It should not be read as a literal child producing multiple families without the unit conversion in Section 5.

2 The conditional household problem

On a maintained old-age branch, elimination of saving gives a coefficient $\rho = 1 + k > 1$ and effective lifetime resources $\widetilde{w}$. The household's conditional problem is:

$$\max_{x,n,h} \; \rho \log x + \alpha \log(h - \kappa n) + \vartheta \log n, \qquad \rho x + \chi n + uh = \widetilde{w}, \quad h \leq \widehat{h}. \tag{4}$$

The same ρ appears in utility and the budget because saving and future consumption move proportionally to usable current goods on this branch.

With interior old-age choices, $\rho = 1 + \beta(1 + \gamma + \omega_B)$ and $\widetilde{w} = y + b + T_t + qT_{t+1}$. When an old renter's housing limit binds, $\rho = 1 + \beta(1 + \omega_B)$ and the fixed future rent must instead be subtracted from resources:

$$\widetilde{w} = y + b + T_t + qT_{t+1} - q^2 r_{t+1} h_R^{\max}. \tag{5}$$

Neither the coefficient nor the future-rent deduction can be borrowed from the interior branch. Saving, tenure, and the other constraints must also remain on the maintained branch.

At a binding current housing limit, the fertility first-order condition is:

$$\frac{\vartheta}{n} = \frac{\chi}{x} + \frac{\alpha\kappa}{s}. \tag{6}$$

An additional child uses both adult consumption and adult space. The young household's willingness to pay for another housing unit is $m_Y = \alpha x / s$; the housing constraint binds strictly when $m_Y > u$.

3 A compensated allocation improvement at fixed fertility

The planner's question is whether one can make existing households better off by moving some housing from an old owner to a young buyer, without changing fertility or relying on the utility of future additional people.

Consider an old owner j with consumption c_j, retained housing h_j, and estate e_j. Suppose both its retention bound and estate-composition bound are slack. Its retention condition equates willingness to pay to user cost:

$$m_O = \frac{\gamma c_j}{h_j} = u_t. \tag{7}$$

Match it with a young buyer i for whom the down-payment limit is the binding housing restriction and $m_Y > u_t$. These are comparisons in units of the consumption good, not interpersonal comparisons of marginal utility levels.

Let $\epsilon > 0$ be the housing transferred and let $C(\epsilon) \geq 0$ be its real cost in date-t goods. Initially assume $C(0) = 0$ and the right derivative $C'(0) = k_c \geq 0$. An exact compensating transfer to the old seller is:

$$L(\epsilon) = c_j \left[\left(\frac{h_j}{h_j - \epsilon} \right)^{\gamma} - 1 \right] - u_t \epsilon. \tag{8}$$

After the sale, keeping its estate fixed leaves the seller $u_t \epsilon$ for consumption; adding $L(\epsilon)$ holds its utility exactly constant. The old retention condition implies $L(0) = L'(0) = 0$ and $L(\epsilon) > 0$ for sufficiently small positive ϵ.

The transaction can be recorded without an unfunded government rebate:

Party	Date t	Date $t+1$
Young buyer	Borrows $(P_t + q\tau^p P_t)\epsilon + L(\epsilon) + C(\epsilon)$; buys title, places $q\tau^p P_t\epsilon$ in a bond reserve, pays compensation and real cost.	Sells the additional unit for $P_{t+1}\epsilon$ and repays the loan; the reserve returns $\tau^p P_t\epsilon$ and pays the tax.
Old seller	Sells ϵ, receives $P_t\epsilon + L$; consumes $c_j(h_j/(h_j - \epsilon))^\gamma$; keeps the estate fixed using the bond and its lower property-tax obligation.	Estate spending is unchanged. The estate no longer sells this unit.
Lenders	Advance the loan at the market bond price.	Receive principal times $1/q$; no loss.
Government	Ownership changes; the taxable stock does not.	Total property-tax receipts and common rebates are unchanged.

The seller invests $qP_{t+1}\epsilon$ of the title receipt to replace the estate's missing sale proceeds and saves $q\tau^p P_t\epsilon$ on its own property-tax reserve. The remainder of the title receipt and tax saving is $u_t\epsilon$, which is available for consumption. The buyer's next-period sale replaces the seller estate's baseline sale. Housing allocation after date t can therefore be held fixed. The baseline permits external goods trade; the bond finances dated expenditures rather than creating consumption resources.

Proposition 1 (Local allocation gain with explicit compensation)**.** *Maintain the stated interior old-owner conditions, the young buyer's positive saving and slack old-age housing and estate constraints, and slackness of its nonfinancial housing bound. Hold fertility, tenure, cohort masses, prices, common rebates, and other households' allocations fixed. If $m_Y - u_t > k_c$, the stated transaction is feasible for all sufficiently small $\epsilon > 0$, leaves the old seller exactly indifferent, and makes the young buyer strictly better off. All other households, estates, and lenders can be left at their baseline allocations or returns.*

Proof. The seller's utility change is exactly zero by (8). Its lower retention relaxes both relevant inequalities, and its unchanged estate is feasible for a small sale. The buyer's young consumption and its baseline old housing and estate can remain fixed. Pay its net financing bill entirely from its old consumption, reducing that consumption by $[u_t\epsilon + L(\epsilon) + C(\epsilon)]/q$. The resale and debt entries in the table give exactly this bill. All strict positivity and slackness conditions persist for a sufficiently small transfer. The young buyer's lifetime utility change is:

$$\Delta U_i = \alpha \log\left(1 + \frac{\epsilon}{s_i}\right) + \beta \log\left(1 - \frac{u_t\epsilon + L(\epsilon) + C(\epsilon)}{qc_{2i}}\right). \tag{9}$$

The interior saving condition gives $c_{2i} = \beta x_i/q$, so the right derivative at zero is $(m_Y - u_t - k_c)/x_i > 0$. This strict derivative proves a positive gain for a sufficiently small finite transfer. Property revenue, next-date housing supply, and other households' bundles are unchanged, and the loan is repaid at the market rate. $\square$

This establishes inefficiency relative to a planner who can extend closing credit. It does not establish constrained inefficiency under the original private credit limit, nor that every old household should sell. The allocation comparison holds fertility fixed, although children enter the definition of the young household's baseline usable space.

A simpler transfer to revisit. Exact household-specific compensation is stronger than necessary for this local result. Choose a uniform per-unit seller premium a with $0 < a < m_Y - u_t - k_c$. Replace $L(\epsilon)$ by $a\epsilon$, financed by the buyer. The old seller now gains to first order by $a\epsilon/c_j$, while the buyer still gains. For several pairs, the same premium works under a uniform positive surplus bound and uniform slackness and curvature bounds. An ordinary market-price sale with zero premium leaves the old seller indifferent only to first order and generally worse off at second order. The uniform premium is a concrete alternative for the author's W4 revisit; it has not been selected as the final policy instrument.

Fixed moving costs. A positive fixed cost $f\mathbf{1}\{\epsilon > 0\}$ violates $C(0+) = 0$. The marginal theorem does not cover it. A sufficient finite-trade test instead requires feasible positive consumption and:

$$u_t\epsilon + L(\epsilon) + C(\epsilon) < qc_{2i}\left[1 - \left(1 + \frac{\epsilon}{s_i}\right)^{-\alpha/\beta}\right]. \tag{10}$$

This is exactly the positivity condition in (9); a fixed move must generate enough total benefit to pay its discrete resource cost. No cost size or functional form is calibrated here.

4 A conditional fertility prediction

Let $\mathcal{R}$ be the lifetime resources left after housing and compensating payments, so that $\rho x + \chi n = \mathcal{R}$. The marginal net payment for additional housing is $p = -\mathrm{d}\mathcal{R}/\mathrm{d}h$. Implicit differentiation gives:

$$\frac{\mathrm{d}n}{\mathrm{d}h} = \frac{\alpha\kappa/s^2 - \chi p/(\rho x^2)}{\Delta}, \qquad \Delta = \frac{\vartheta}{n^2} + \frac{\chi^2}{\rho x^2} + \frac{\alpha\kappa^2}{s^2} > 0. \tag{11}$$

More space raises fertility if its benefit dominates the resources used to pay for the additional housing. The payment includes compensation and real costs when those are charged to the young household.

Proposition 2 (Fertility and the net payment)**.** *On the maintained reduced branch, fertility increases locally exactly when $\rho\kappa m_Y^2 > \alpha\chi p$. If $\kappa u/\chi > \alpha/\rho$ and the young household is housing constrained, fertility increases for any marginal payment $p < m_Y$ that leaves it better off. This includes a pure relaxation of the cap with $p = u$ and the compensated marginal allocation whenever $p = u + k_c < m_Y$.*

Proof. Multiply the numerator of (11) by the positive number ρx^2 and use $m_Y = \alpha x/s$. For the sufficient condition, $m_Y \geq u$ implies $\rho\kappa m_Y > \alpha\chi$, hence $\rho\kappa m_Y^2 > \alpha\chi p$ whenever $p < m_Y$. □

The result contains no multiplier, but its sufficient condition still contains the housing user cost. The next result removes that price in a stationary specialization. It does not remove the branch and strict-binding qualifications.

4.1 A sufficient condition entirely in primitives

At stationary prices, set $d = 1 - q + q\tau^p > 0$, so $u = dP$. If the down-payment limit alone binds, $\widehat{h} = b/[(1-\phi)P]$. The lifetime share spent on housing at this limit is independent of P when $\widetilde{w}$ is exogenous:

$$a_h = \frac{u\widehat{h}}{\widetilde{w}} = \frac{db}{(1-\phi)\widetilde{w}}. \tag{12}$$

For interior old choices, zero property tax, and the corresponding zero common rebate, $\widetilde{w} = y + b$. This is an explicitly labeled specialization of the property-tax model, not an assumption that its endogenous rebate is fixed in general equilibrium.

Proposition 3 (A price-free sufficient restriction)**.** *Maintain the reduced problem, stationary prices, strictly binding down-payment limit, and the other branch conditions. If $\widetilde{w}$ is exogenous and*

$$\frac{db}{(1-\phi)\widetilde{w}} \geq \frac{\alpha}{\rho + \alpha}, \tag{13}$$

then $\kappa u/\chi > \alpha/\rho$. Consequently a marginal cap relaxation at fixed prices and resources raises fertility. More generally, any local increase in housing with net marginal payment $p < m_Y$ raises fertility on the same branch. In the zero-property-tax, interior-old specialization the left side is $(1-q)b/[(1-\phi)(y+b)]$, and $\rho = 1 + \beta(1+\gamma+\omega_B)$: the restriction contains neither prices nor multipliers.

Proof. Let $D = \rho + \alpha + \vartheta$ and $v = \kappa u/\chi$. Without the current cap, the conditional problem spends the following fraction of resources on housing:

$$g(v) = \frac{\alpha + \vartheta v/(1+v)}{D}. \tag{14}$$

Strict concavity implies that the current cap binds strictly exactly when $a_h < g(v)$ in this reduced problem. The function g is strictly increasing, and direct substitution gives $g(\alpha/\rho) = \alpha/(\rho+\alpha)$. Therefore (13) and strict binding imply $g(v) > g(\alpha/\rho)$ and hence $v > \alpha/\rho$. Proposition 2 supplies the fertility implication. □

The restriction is not vacuous: a strictly binding cap can satisfy it when $\alpha/(\rho+\alpha) \le a_h < (\alpha+\vartheta)/D$. The interval has positive length. It is nevertheless a substantive restriction: it does not cover arbitrarily low liquidity relative to lifetime resources. Within the reduced problem, smaller a_h cannot give a uniform positive cap effect at every price where the cap binds. Near a slack-to-binding boundary with $v < \alpha/\rho$, the derivative has the opposite sign. If the cap already binds as $v \downarrow 0$, the space term in the numerator is of order u^2 and the resource term is negative of order u. These observations restrict uniformity over prices; they are not a claim that every such price is an equilibrium or satisfies every other branch inequality.

A constructed example also satisfies the underlying owner-branch restrictions. Set $q = 0.8$, $\phi = 0.7$, $P = 4$, $y = 4.8$, $b = 2.4$, and $\tau^p = T = 0$. Let $\beta = 0.8$, $\gamma = 0.5$, $\omega_B = 3$, $\alpha = \chi = \kappa = 1$, and $\vartheta = 2$. Then $\rho = 4.6$, $u = 0.8$, $h = 2$, and $x = s = n = 1$. The share $a_h = 2/9$ exceeds $\alpha/(\rho+\alpha) = 5/28$, while $m_Y = 1 > u$. Saving is 2.8, old consumption is 1, old housing is 0.625, and the estate is $3.75 > 4(0.625)$. Thus positive saving and the two old-age slackness conditions are consistent with the restriction. The cap derivative is $19/74 > 0$. This is a conditional household example, not a complete equilibrium construction.

Endogenous rebates, transition appreciation, tenure changes, and old-age branch changes require additional work. The fully primitive specialization does not prove a positive aggregate equilibrium response to a housing policy. For a finite reform, the local sign must continue to hold along the relevant household path; no such global sign has been established here.

4.2 Two checks that must accompany the statement

Welfare alone does not sign fertility. Set $\rho = 2$, $\alpha = \chi = x = s = n = 1$, $\kappa = 0.1$, $\vartheta = 1.1$, and $u = 0.5$. Then $h = 1.1$, $\widetilde{w} = 3.55$, $m_Y = 1$, and $\Delta = 1.61$. At $p = u$, the household is better off with more housing but $\mathrm{d}n/\mathrm{d}h = -15/161$. If the added space is provided at zero marginal net payment, the same conditional derivative is $10/161 > 0$. This is a constructed household example, not a calibration or an equilibrium counterfactual.

A common rental limit changes future resources. If both the young and old rental caps bind, raising their common limit changes current space and the old household's committed rent. At fixed prices and transfers its total derivative is:

$$\frac{\mathrm{d}n}{\mathrm{d}h_R^{\max}} = \frac{\partial n}{\partial \widehat{h}}\bigg|_{\widetilde{w}} - q^2 r_{t+1} \frac{\partial n}{\partial \widetilde{w}}, \qquad \frac{\partial n}{\partial \widetilde{w}} = \frac{\chi}{\rho x^2 \Delta} > 0. \tag{15}$$

Reporting the first term alone overstates the fertility response to the common cap change on this branch.

5 Population comparisons and units

For two paths starting with the same young cohort, iteration of the cohort law gives an exact transition comparison:

$$\frac{Y_t^1}{Y_t^0} = \prod_{s=0}^{t-1} \frac{\bar{n}_s^1}{\bar{n}_s^0}. \tag{16}$$

Thus cumulative relative fertility determines relative young-cohort size; the old-cohort comparison follows with a one-period lag. This identity does not prove that a policy raises fertility in every period or that either path converges.

At a positive stationary population, fertility must be $\bar{n}^* = 1/\nu$ and $Y^* = O^*$. Define $\bar{h}_{\text{hh}}^* = (\bar{h}_Y^* + \bar{h}_O^*)/2$, housing per adult household. Fixed-stock housing clearing gives:

$$N_{\text{hh}}^* = Y^* + O^* = \frac{\bar{H}}{\bar{h}_{\text{hh}}^*}. \tag{17}$$

Two policies can therefore have the same replacement fertility but different terminal population levels. A larger level requires less housing per household at the terminal allocation when supply is fixed.

Child-unit correction. Let $n^L = a_n n$ be literal children per household. The equivalent demographic coefficient is $\nu^L = \nu/a_n$, so replacement literal fertility is a_n/ν. The equivalent household requirements are $\chi^L = \chi/a_n$ and $\kappa^L = \kappa/a_n$; replacing $\log n$ by $\log n^L$ changes utility only by a choice-independent constant. Thus the rescaling preserves behavior when all relevant objects are transformed together. With $\nu = 2$, replacement is $n = 0.5$. A conversion $a_n = 2.1$ gives 1.05 literal children, not 2.1; to map 0.5 to 2.1 requires $a_n = 4.2$ and $\nu^L = 1/2.1$. The algebraic correction does not calibrate the toy or change quantitative model units.

Households and resident persons. N_{hh} counts adult household units. Converting it to resident persons requires the number of adults per household and the timing and residence of children. If these are fixed coefficients a_Y, a_O and children are counted as dependents in their birth period, a consistent illustration is $N_{\text{persons,t}} = a_Y Y_t + a_O O_t + a_n \bar{n}_t Y_t$. At a positive steady state the ratio $N_{\text{persons}}^*/N_{\text{hh}}^*$ is $(a_Y + a_O + a_n/\nu)/2$. Fixed common coefficients preserve the ordering of terminal household and person counts. Their empirical values and the within-period counting convention are not selected by this illustration.

A stationary housing-supply extension. If the terminal stock is $S(P^*) > 0$, the scale identity becomes $N_{\text{hh}}^* = S(P^*)/\bar{h}_{\text{hh}}^*$. Hence the exact comparison is:

$$\log \frac{N_{\text{hh},1}^*}{N_{\text{hh},0}^*} = \log \frac{S(P_1^*)}{S(P_0^*)} - \log \frac{\bar{h}_{\text{hh},1}^*}{\bar{h}_{\text{hh},0}^*}. \tag{18}$$

When the only fiscal transfer is the property-tax rebate, $T^* = q\tau^p P^* \bar{h}_{\text{hh}}^*$, the housing-stock scale cancels from the stationary household and fiscal equations. Supply may then change terminal population without changing their stationary price roots. This observation requires supplier profits, land income, and construction financing not to add new household-income or fiscal effects. A supply curve alone does not define the transition law or its resource costs.

6 Transition proof and amendment status

The broad transition question remains unfinished at this checkpoint. Removing the gains tax removes its particular price kink, but does not establish global existence, convergence, uniqueness, or the sign of a policy response. The existing finite-horizon construction and smooth local transition proposal must be reconciled with the tax-free core and checked at the relevant zero-shock state. A larger theorem must state its domain, branch or aggregation regularity, admissibility, and terminal selection conditions explicitly.

The remaining overnight work will assess a continuation argument beyond an arbitrarily small reform, identify the actual obstacles, and separate any theorem from numerical illustration. The existing calibrated quantitative model and its policy experiments are outside this theory amendment task. The companion work record retains all 18 decision entries and the 14 review repairs. The precise compensation instrument, real cost, and resident-person interpretation remain concrete points for the author's review.